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5G, Edge Computing, and APIs: **What CIOs and CTOs Need to Know**

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Executive summary

In the previous decade, telcos generated a total shareholder return of 5% compared to 30% generated by OTT (over-the-top) streaming services and 17% by IT infrastructure companies. Edge players, such as businesses developing apps, providing streaming and other digital services, and building infrastructure, have capitalized on telco investments in successive technology waves even as telcos lagged.

Telcos have invested approximately $1 trillion in network upgrades since 2018 but are still at risk of losing a chance to monetize their 5G investments. They must improve data throughput, reduce latency, and create services and applications to move beyond the 5G value proposition of enhanced connectivity. To avoid ceding value creation opportunities to other entities in the ecosystem, such as cloud providers, API aggregators, and innovative consumer product companies, telcos must leverage the combined capabilities of 5G, edge computing, and APIs.

APIs Unlocking Value for Telcos

A McKinsey report estimates that telco operators can realize $100-$300 billion in connectivity and edge-computing-related revenue through network APIs, potentially generating an additional $10-30 billion from APIs themselves by 2031. Enterprises can leverage APIs to seamlessly tap into 5G capabilities for numerous use cases such as credit card fraud prevention, self-driving automobiles, and entertainment services. Collaboration with enterprise and independent developers

Monetizing 5G requires telcos to open the network features and capabilities to developers through APIs, enabling them to build innovative applications for consumers and enterprises. The shift from 'connectivity pipes' to programmable platforms will offer new revenue streams for telcos while supercharging innovation in the app economy.

Network APIs will enable developers to build applications that utilize network resources and add new functionalities at scale, efficiently bringing new products and services to the market. Telcos must develop ongoing relationships with enterprise and independent developers, an essential source in the innovation ecosystem. The initial efforts of telcos to engage developers have not been successful because of a lack of cross-network interoperability, which developers need to change to find a value proposition appealing to the developer community.

Global Interoperability

Telco's efforts to foster collaboration within the broader telecom industry are gaining momentum with the formation of CAMARA, which is an open-source project focused on developing standardized network APIs. The initiative between GSMA (Global System for Mobile Communication) and Linux Foundation creates common definitions for APIs, enabling developers to use a single piece of code to access 5G capabilities across networks.

Open Gateway, another initiative of GSMA, identifies the most marketable APIs, and early use cases from this initiative are already emerging. Twelve leading global telecom operators and Ericsson have formed an equal joint venture to sell network APIs, taking the Open Gateway initiative to the next level. The network APIs will be based on CAMARA APIs and offered to developer platforms, including hyper scalers, system integrators, and ISVs (independent software vendors), to enable them to create applications with ease.

Customized Enterprise Solutions

Telcos can offer enterprise-customized solutions and services leveraging advanced 5G capabilities such as location-based services and authentication. The global chemical and industrial company, BASF, uses Vonage APIs to connect 30,000 farmers across 20 nations via WhatsApp. The company has used geo-location-based analytics to empower farmers with information and resources to help them improve productivity and realize better value for their agricultural produce.

Similarly, Telefonica Vivo Brazil provided customized anti-fraud solutions to financial services companies to enable them to validate their customer identities across the value chain from account creation to transaction. The telco leveraged its network capabilities through APIs to help financial firms reduce fraud and monetize their 5G network capabilities.

Premium Consumer Services

Quality-of-Service on Demand (QoD) APIs provide users with predictable network conditions that significantly enhance their experiences for specific applications or use-case. Telcos can charge users premiums, which can be bundled into the service offering or dynamically controlled by users.

Telcos can use QoD APIs to provide differentiated premium services, enhancing the user experience and increasing customer loyalty and satisfaction. The demand for such premium services exists among as customer segments such as gamers, who are willing to pay for 5G boosters that can temporarily improve the gaming or video streaming experience. Research has found that smartphone users are willing to pay an average premium of 11% for quality services.

Telcos can first prioritize simple APIs related to location and fraud prevention that are relevant to several industries and use cases. Subsequently, QoD, network slicing, and security APIs offer monetization opportunities on 5G networks. After creating valuable proof points with the two classes of APIs, telcos can focus on APIs that will help them access edge computing markets.

Edge Computing: 5G Success Driver

With edge computing, organizations can experience a drop in latency, bandwidth increase, and more sovereignty over their data. In a survey, 55% of telecom executives said their primary objective of using edge computing was to improve network efficiency. Another 21% cited enabling new use cases for business and 18% favored new use cases for consumers.

With computing workloads located closer to end users, edge computing enables real-time data processing that unlocks different use cases across industries. It provides a platform for next-level automation leveraging Machine Learning (ML) and Artificial Intelligence (AI). On-premises edge computing minimizes the risk of data being intercepted in transit, improving security and privacy.

The low latency from edge offers clear advantages in gaming, offering telcos monetization opportunities. There are more than 100 use cases across manufacturing, banking and insurance, healthcare, energy and materials, retail, and public sector utilities that give telcos new opportunities to drive value from their 5G network investments.

With growing demands, consumers expect seamless digital interactions and innovative solutions for everyday activities related to banking, retail, gaming, and entertainment. With the number of Internet of Things (IoT) devices expected to grow to an estimated 20.1 billion by 2025, the demand for connectivity from both B2C and B2B segments will intensify. Telcos have an opportunity to capitalize on these evolving market expectations by leveraging edge computing and APIs. Innovative 5G services and solutions hold the key to driving new revenue growth, generating higher return on investments, and creating more shareholder value.

About us

Torry Harris Integration Solutions (THIS) enables digital ecosystems through adoption and use of API driven digital opportunities. We bring 25+ years of focused integration experience with impeccable, long standing references in enabling digital transformation for enterprises. We are headquartered in New Jersey, USA with offices in UK, Ireland, France, Germany, Spain, UAE and India.

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